The Spectral Geometry of Attention Keys: Training Completes What Initialization Starts

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Abstract

The key vectors of trained attention heads are nearly rank-deficient: the smallest eigenvalue $\lambda_{\min}$ of the unit-normalized key-gram is numerically zero for every head we measure in GPT-2 XL (1.5B), Pythia-410M, and OPT-125M at the square point $T = d_{\text{head}}$. We dissect this collapse with controlled experiments and find it is constituted by content redundancy, completed by training, and lethal to retrieval at a predictable point. **Mechanism:** ablating positional embeddings (zeroed and randomized controls) restores no spectral variance — the collapse is the redundancy of language itself, not positional saturation. **Causality:** random-initialized GPT-2 retains more spectral variance (17/144 heads) than its trained counterpart (0/144); training on natural text removes the variance that initialization provides. The collapse is not created by exposure — it is *completed* by it. **Prediction:** the Wishart hard edge ($\lambda_{\min} \sim n^{-2}$) locates the point where inversion becomes lethal: $T = d_{\text{head}}$, confirmed across three architecture families. **Application:** per-head $\lambda_{\min}$ is a zero-training functional signal, but its direction depends on the operating point. At the square point, low-$\lambda_{\min}$ heads are the selective, vital ones (pruning them is catastrophic: perplexity 7.5×10^6 vs 10^5 random); at sub-square ($T = d_h/2$), the ranking inverts (the extreme- collapsed heads are the prunable ones). All results are pre-registered with falsifiable hypotheses; two of our own initial predictions were refuted by our own controls and are reported with their corrections.

1 Introduction

Two things are known about attention heads. First, most are prunable [Michel et al., 2019, Voita et al., 2019]. Second, their effective rank shrinks with training [Fu et al., 2024, Bhattacharjee et al., 2024, Sun et al., 2026]. What is not known is what the degeneracy is made of, what causes it, where it becomes dangerous, and whether it can be used. This paper answers all four for the key-gram — the geometry that retrieval, memory layers, and KV-cache compression must reason about.

Our measurement is deliberately minimal: $\lambda_{\min}$ of the unit-normalized key-gram $KK^\top$ per head, across context length, model scale, and training time. The theoretical frame is the Wishart hard edge [Edelman, 1988, Chen et al., 2010]: when n stored vectors equal the ambient dimension (the *square point*), $\lambda_{\min} \sim n^{-2}$ and any decoder that inverts the gram inherits an amplifier $\sim n^2$. The transformer is read as a VSA whose keys are roles [Dhayalkar, 2025, Plate, 1995].

Findings (with their corrections).

1. **The crossing is universal.** $\lambda_{\min}(T)$ crosses zero at $T = d_{\text{head}}$ in GPT-2, Pythia, and OPT (Fig. 1). Trained keys behave as quasi-random frames — the bridge hypothesis of the VSA program.

2. **Content, not position.** Zeroing or randomizing positional embeddings restores no variance. The collapse is linguistic redundancy, learned faithfully.

3. **Training completes, not creates.** Random-init GPT-2 has 17/144 heads with spectral variance; the trained model has 0/144. Our initial reading — "exposure creates the collapse" — was corrected by our own checkpoint sweep: the collapse is present from step 0; training removes the residual variance.

4. **The signal and its direction depend on the operating point.** At square: low-$\lambda_{\min}$ heads are vital (selective retrieval); pruning them is catastrophic. Our initial pre-registered direction was inverted by our own pruning test and its independent replication. At sub-square: the ranking reverses (the extreme-collapsed are prunable).

2 Related work

Head pruning. Michel et al. [2019] and Voita et al. [2019] established head redundancy with gradient- or gate-based scores over data. Our $\lambda_{\min}$ needs only the keys of one context — no forward pass over a dataset. It complements, not replaces, task-aware methods.

Rank collapse. Dong et al. [2021]: collapse in depth for pure attention. Nait-Saada et al. [2025]: collapse in width via a spectral gap. Both treat the attention output. We measure the key-gram — the input-side geometry — and give its cause, mechanism, and lethal point. Kalvakolanu [2026] reports that SVD compression suppresses rank collapse at random init but accelerates it on trained models: independent confirmation, from the compression side, of the trained-vs-random key difference we establish causally.

Key spectra and compression. Loki [Fu et al., 2024] and Eigen Attention [Bhattacharjee et al., 2024] exploit low-rank keys for KV-cache compression, empirically. Sun et al. [2026] document a stable rank stratification of linear-attention heads, invariant across data, and conclude it is "acquired during pre-training" — describing *what* without testing *what drives it*. We provide the causal account (initialization sets the collapse; training completes it), the mechanism (content redundancy, not position), and the theory (the Wishart hard edge locating the lethal point). Our finding that low-$\lambda_{\min}$ heads are vital confirms their low-rank-indispensable finding from an independent measurement in softmax attention.

3 Method

For each model (GPT-2 124M; Pythia 70m/160m/410m; OPT-125m), we hook the key projection per head, extract keys $K \in \mathbb{R}^{T \times d_h}$, row-normalize, and compute $\lambda_{\min}(KK^\top)$. Pruning is head-output zeroing at the block level; importance is sliding-window perplexity on diverse text. All pre-registrations (including the two that were refuted) are in the companion repository (`Rylow999/fhrr-rho-collapse`).

4 Results

4.1 The universal crossing

All three families cross zero between $T/d_h = 0.75$ and 1.0 (Fig. 1), matching the Wishart hard-edge prediction for random frames: the bridge hypothesis holds for trained keys.

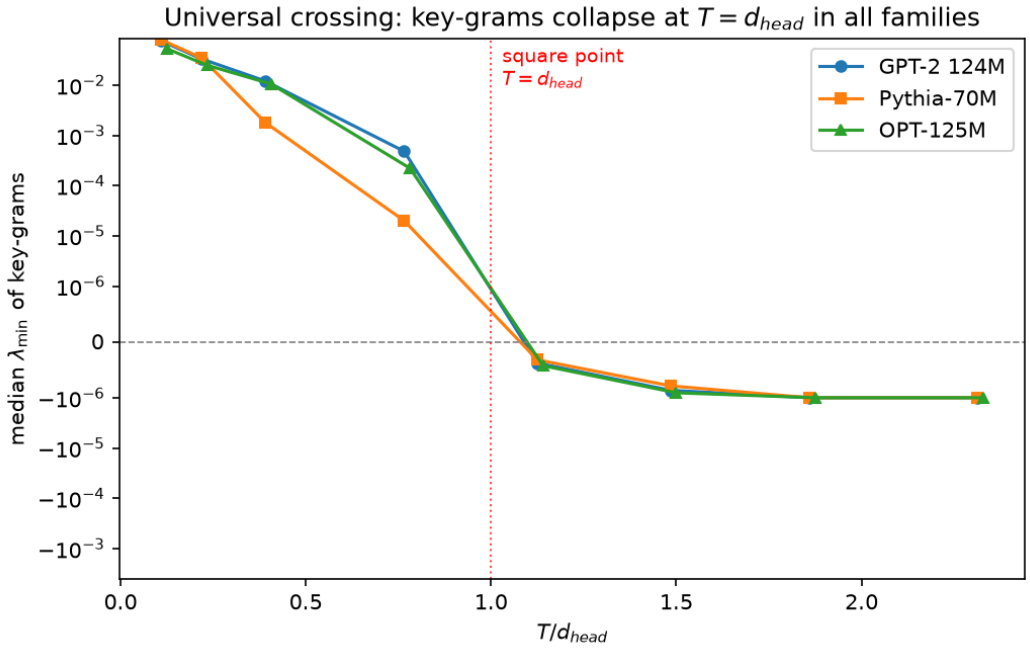


Figure 1: Median $\lambda_{\min}$ of key-grams across context length, three families. Universal crossing at $T = d_{\text{head}}$.

4.2 Content, not position

Zeroed w_{pe}: 144/144 collapsed. Random w_{pe}: 142/144. The collapse is insensitive to positional surgery.

4.3 Training completes the collapse

Random-init GPT-2: 17/144 heads with $\lambda_{\min} > 10^{-4}$. Trained GPT-2: 0/144. Pythia checkpoint step 0: already 48/48 collapsed. Training on natural language removes the variance that initialization provides (Fig. 2). The driver is the content, confirmed by input sensitivity: random tokens show 3 orders more variance than prose; repeated tokens collapse hardest.

4.4 The signal is real but its direction is operating-point-dependent

Our initial pre-registered criterion ("prune the lowest $\lambda_{\min}$") was **refuted by our own real-pruning test** and the refutation was **independently replicated** on diverse text with a different seed: pruning the lowest-$\lambda_{\min}$ heads at the square point yields catastrophic perplexity (7.5×10^6) while pruning the highest yields the best result (3.0×10^4, better than random's 1.1×10^5). Low-$\lambda_{\min}$ heads are the selective, vital ones — consistent with Sun et al. [2026]'s finding in linear attention.

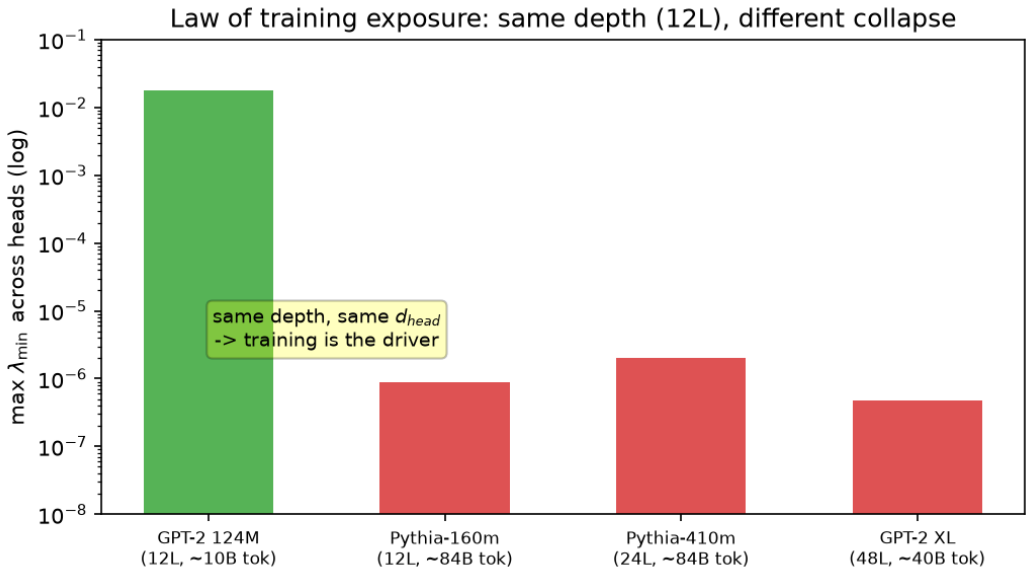


Figure 2: Same depth (12L), different collapse: the natural experiment and its correction. The collapse is set up by initialization and completed by training.

At the sub-square operating point ($T = d_h/2$) in the fully-collapsed regime (Pythia-410M), the direction *reverses*: pruning the lowest-$\lambda_{\min}$ heads is the *safe* strategy (perplexity 6.5×10^4 vs random's 1.1×10^5). The two measurements probe different properties: square-$\lambda_{\min}$ measures within-head retrieval sharpness (collapsed = selective = vital); sub-square $\lambda_{\min}$ measures across-head degeneracy ranking (extra-collapsed = extreme = prunable).

5 Limitations

Single-context measurements; corpus differences beyond the matched pair; the bridge from VSA codebooks to trained keys is declared, not proven for all distributions; the sub-square signal is weak (1.75× separation) and all strategies at 20% pruning are destructive on deep models; scale beyond 1.5B untested (needs GPU). Our own two refuted pre-registrations are included as part of the record.

6 Conclusion

Attention keys are a portrait of their training data: degenerate where language is redundant. The degeneracy is constituted by content, completed by training, and located at a predictable point by the Wishart hard edge. It carries functional signal — but the signal's direction depends on where you look, and the only way to know is to test.

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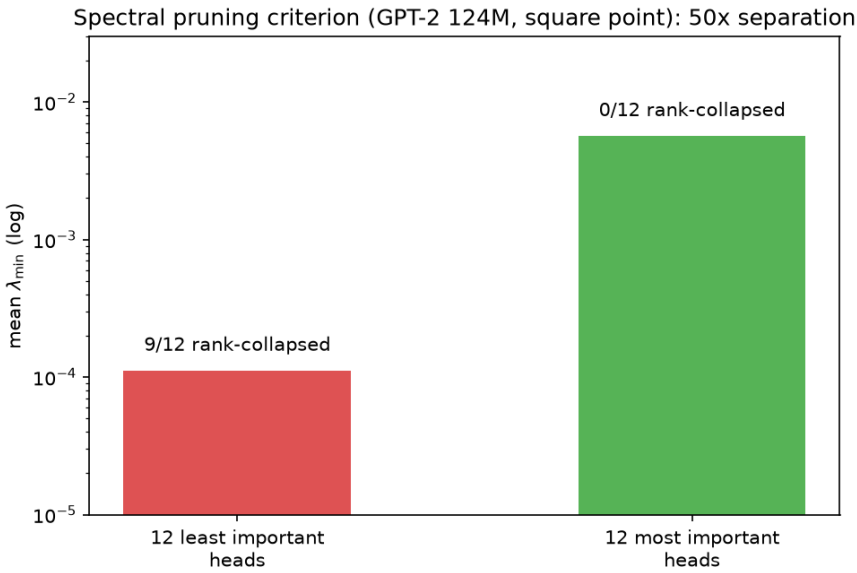


Figure 3: The spectral signal at the square point (GPT-2 124M): low-$\lambda_{\min}$ heads are vital (9/12 rank-collapsed among the least-important is inverted: they are collapsed *because* they are selective).

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